



CHAPTER 02

Variables and Datatype



Variables

A **variable** is the name given to a memory location in a program. For example.

```
a = 30      # variables = container to store a value
b = "harry" # keywords = reserved words in python
c = 71.22   # identifiers = class/function/variable name
```

Variable

Name

Memory
Location

Value

Think of a
variable as
a container!



Data Types

Primarily these are the following **data types** in Python:

1. Integers
2. Floating point numbers
3. Strings
4. Booleans
5. None

Data Type	Example
Integers	25, -7, 100
Floating point numbers	3.14, -0.01, 2.0
Strings	"hello", 'Python', "123"
Booleans	True, False
None	None

Different types
of data help
us store
different kinds
of information.

Python is a fantastic language that **automatically identifies** the type of data for us.

```
a = 71      # identifies a as class <int>
b = 88.44   # identifies b as class <float>
name = "harry" # identifies name as class <str>
```

a is of type
int

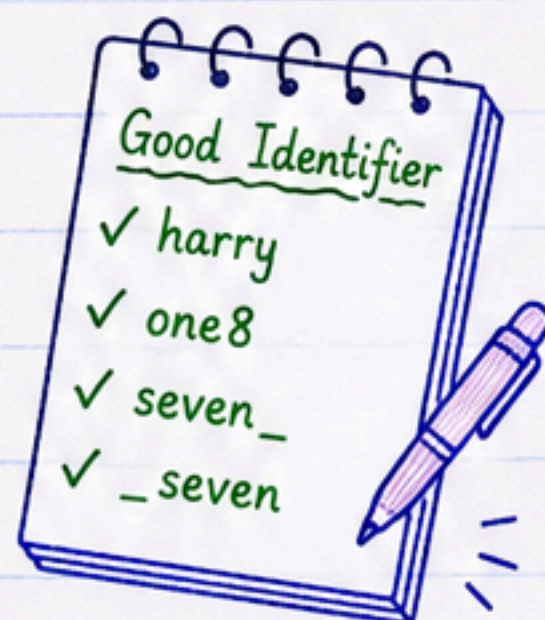
b is of type
float

name is of type
str



Rules for Choosing an Identifier

- A variable name can contain **alphabets**, **digits**, and **underscores**.
- A variable name can only start with an **alphabet** and **underscores**.
- A variable name **can't start with a digit**.
- **No white space** is allowed to be used inside a variable name.



Examples of a few variable names are: **harry**, **one8**, **seven_**, **_seven** etc

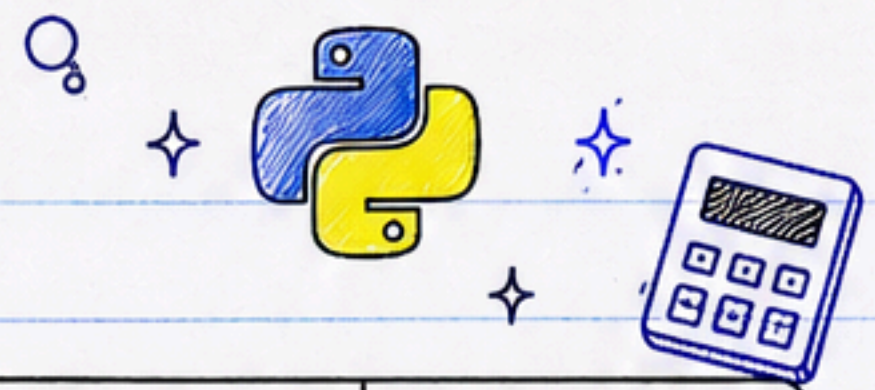


Keep your variable
names meaningful!





Operators in Python



Following are some common operators in python:

1. Arithmetic operators: +, -, *, / etc.
2. Assignment operators: =, +=, -= etc.
3. Comparison operators: ==, >, >=, <, != etc.
4. Logical operators: and, or, not.

Type of Operator	Examples
Arithmetic	+, -, *, /
Assignment	=, +=, -=
Comparison	==, >, >=, <, !=
Logical	and, or, not

type() Function and Typecasting

type() function is used to find the data type of a given variable in python.

```
a = 31
type(a)      # class <int>

b = "31"
type(b)      # class <str>
```

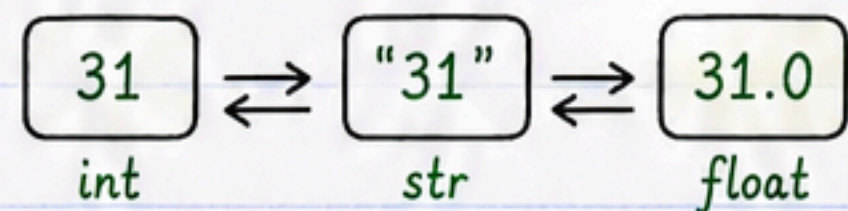


type() tells us what kind of data a variable holds.



- A number can be converted into a string and vice versa (if possible)
- There are many functions to convert one data type into another.

```
str(31)      # integer to string conversion
int("32")    # string to integer conversion
float(32)    # integer to float conversion
... and so on.
```



Different types, same value!



- Here "31" is a string literal and 31 a numeric literal.



Input() Function

This function allows the user to take input from the keyboard as a string.



```
a = input("enter name") # if a is "harry", the user entered harry
```

enter name: harry_

It is important to note that the output of input is always a string (even if a number is entered).



input() always returns a string!



Practice makes perfect!



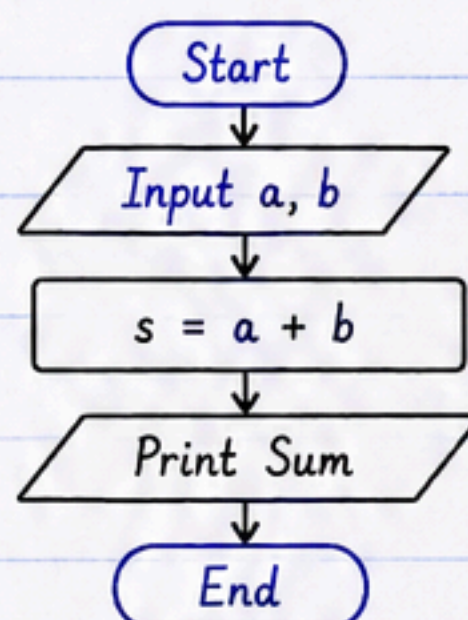


Practice Set ☆ ☆



1. Write a python program to add two numbers.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
s = a + b
print("Sum =", s)
```

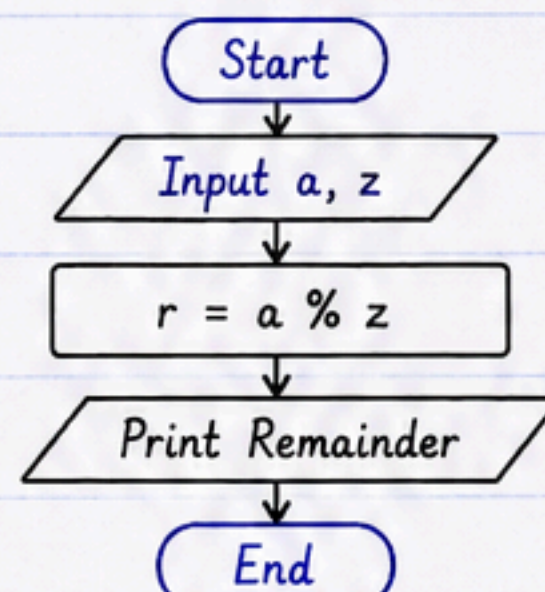


+ is used to add numbers.



2. Write a python program to find remainder when a number is divided by z.

```
a = int(input("Enter a number: "))
z = int(input("Enter z: "))
r = a % z # % gives remainder
print("Remainder =", r)
```

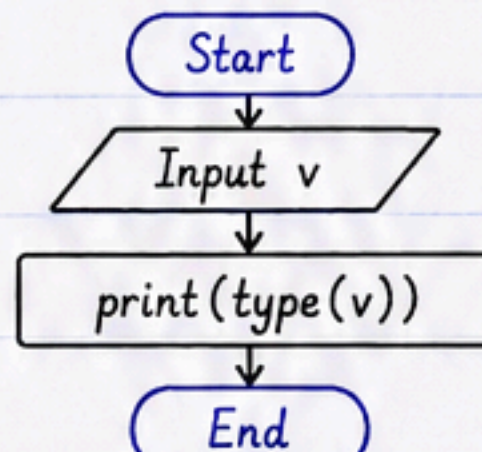


% is the modulus operator.



3. Check the type of variable assigned using input() function.

```
v = input("Enter something: ")
print("Type of v is:", type(v))
```



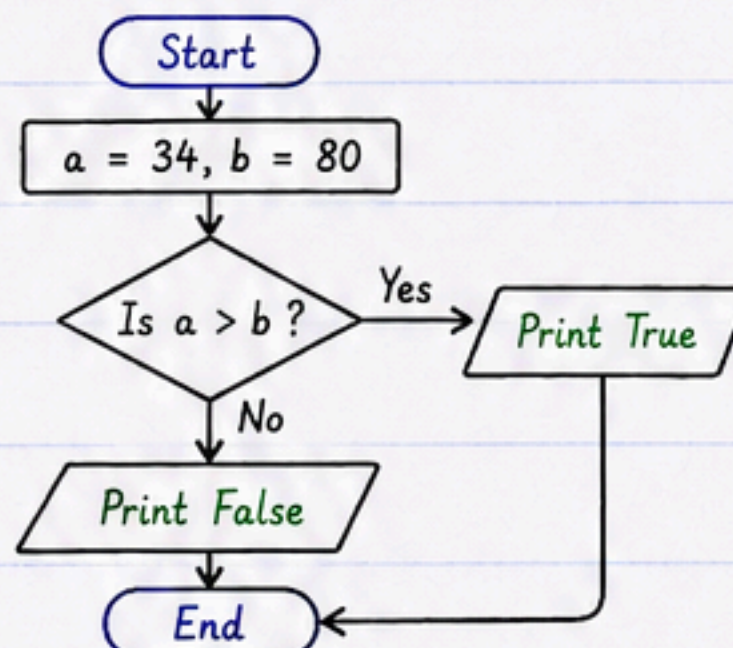
input() always takes input as a string.



4. Use comparison operator to find out whether 'a' given variable is greater than 'b' or not.

Take a = 34 and b = 80

```
a = 34
b = 80
print("Is a greater than b?", a > b)
```

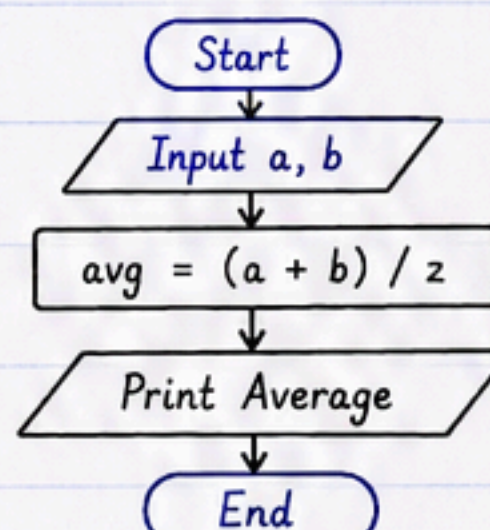


> is used to check greater than.

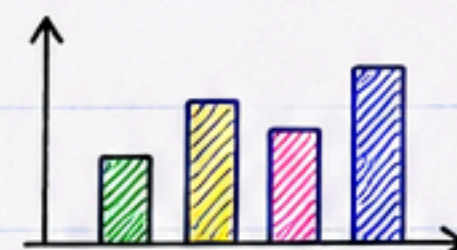


5. Write a python program to find an average of two numbers entered by the user.

```
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
avg = (a + b) / 2
print("Average =", avg)
```

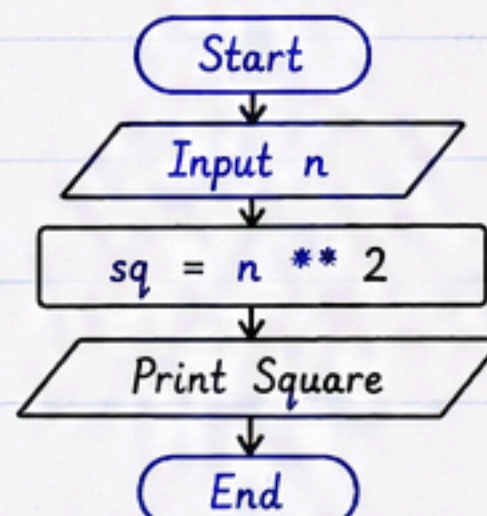


Average = (sum of numbers) / 2



6. Write a python program to calculate the square of a number entered by the user.

```
n = float(input("Enter a number: "))
sq = n ** 2 # ** is power operator
print("Square =", sq)
```



** is used to calculate power.

$$2^2 = 4$$



Keep Practicing
Keep Coding!

